

Technologies to understand cell–microenvironment interactions

Cells continuously interact with neighboring cells and their physical microenvironment, and these interactions shape cell behavior and fate. To understand these interactions, we develop technologies that connect microenvironmental cues with cellular responses. We developed sGRAPHIC, a proximity-labeling approach to investigate molecular responses associated with cell–cell interactions (Nat. Commun., 2023). We also developed MECH-seq, which combines controlled mechanical confinement with molecular profiling to examine how cancer cells respond and adapt to their physical environment. Building on these approaches, we are now extending our efforts toward spatiotemporal analysis to capture how cellular interactions and molecular states change over time. Together, these technologies provide complementary perspectives on cell–microenvironment interactions.